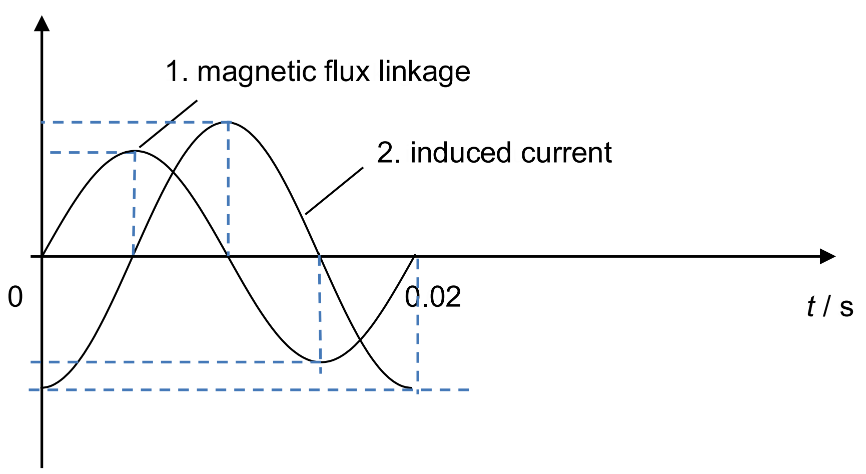
4	**(a)**	Period or time for 1 full cycle = 1/f time to rotate 1° = $\frac{1}{360}\left(\frac{1}{f}\right)=\frac{1}{360}\left(\frac{1}{50}\right)=5.56\times10^{-5}$ s OR since it rotates 50 x 360 = 18000° in 1 s, time taken to rotate 1° = 1/18000 = 5.6 x 10^{-5} s	1
	(b)	Change in flux linkage = $\Delta\Phi$ = Δ(NBA) = NA$\Delta B_{\perp}$ = 38 (0.29)(2.0)(1.2) sin 1.0° - 0 = 0.46 Wb turns	 1 1
	(c)	e.m.f. = $\Delta\Phi$/ Δt = 0.46 / 5.6 x 10^{-5} = 8200 V = 8.2 k V	1
	(d)	The direction of the current induced in side AD is **A to D.** As the coil starts turning, induced emf in the coil is proportional to the rate of increase in flux linkage, by Faraday's Law. **By Lenz's Law, induced current in the coil will be in the direction A to D to produce a leftward magnetic field to oppose the increase of flux linkage to the right.** (The direction of current is predicted using right hand grip rule.)	 1 1
	(e)	 -1 mark for wrong shape of graph or not labelling the period.	1 1